H_C diagonalisation and the analytic coefficients of H_J: transcription of the symbolic results

This note transcribes the analytical output of the Wolfram notebook `hj_analytic_coefficients.nb`. For each $\{k,-k\}$ sector of the periodic SSH photonic chain the capacitive Nambu block H_C is diagonalised by a paraunitary transform $T = T_{\rm rot}\,T_{\rm sq}$ (a phased sublattice rotation followed by a two–mode squeeze); the same T is then applied to the Josephson block H_J and the closed forms of every coefficient of the transformed block $\tilde{H}_J$ are read off. Sections 5–6 truncate the cavity–dressed problem to a two–level system and obtain the Zak phase of the hybridised chain / resonator–photon band analytically. All symbolic results were cross–checked against the numerical implementation `PeriodicChainSW` (see §7).

1 Conventions

Per $\{k,-k\}$ sector, in the Nambu basis

$$\Phi = \big(b_{kA},\, b_{kB},\, b_{-kA},\, b_{-kB},\, b^\dagger_{kA},\, b^\dagger_{kB},\, b^\dagger_{-kA},\, b^\dagger_{-kB}\big)^\top,$$

the "honest" quadratic forms are $H_X = \tfrac{1}{2}\,\Phi^\dagger \mathcal{H}_X\,\Phi$ with 8×8 Hermitian, para–Hermitian matrices $\mathcal{H}_C$, $\mathcal{H}_J$. Physical mode frequencies and coefficients carry the Nambu prefactor 2.

Symbols.

$$\delta_{C,k} = g_{c1} + g_{c2}\,e^{-ik}, \qquad \delta_{J,k} = g_{J1} + g_{J2}\,e^{-ik} \ \text{(generic complex)},$$

$$\kappa_k = |\delta_{C,k}| = \sqrt{g_{c1}^2 + g_{c2}^2 + 2g_{c1}g_{c2}\cos k} \in (0,\tfrac{1}{2}), \qquad g_J = g_{J1} + g_{J2} \ \text{(real)},$$

$$\varphi_k = \arg\delta_{C,k}, \qquad \bar{\delta}_{C,k},\ \bar{\delta}_{J,k} = \text{complex conjugates}.$$

The $-k$ blocks use $\delta_{C,-k} = \bar{\delta}_{C,k}$ and $\delta_{J,-k} = \bar{\delta}_{J,k}$.

Write $h(x,\delta) = \left(\begin{smallmatrix} x & \delta \\ \bar{\delta} & x \end{smallmatrix}\right)$ for the Bloch 2×2.

Capacitive block. With 2×2 sub–blocks (rows/cols A,B),

$$\mathcal{H}_C = \begin{pmatrix} \mathcal{N}_C & \mathcal{P}_C \\ \mathcal{P}_C^* & \mathcal{N}_C^* \end{pmatrix}, \qquad \mathcal{N}_C = \tfrac{1}{2}\,\mathrm{diag}\big(h(1,\delta_{C,k}),\ h(1,\bar{\delta}_{C,k})\big),$$

and $\mathcal{P}_C$ has the nonzero entries $(\mathcal{P}_C)_{14} = (\mathcal{P}_C)_{41} = -\delta_{C,k}/2$, $(\mathcal{P}_C)_{23} = (\mathcal{P}_C)_{32} = -\bar{\delta}_{C,k}/2$.

Josephson block. $\mathcal{H}_J$ has the same layout with

$$\mathcal{N}_J = \tfrac{1}{2}\,\mathrm{diag}\big(h(g_J,-\delta_{J,k}),\ h(g_J,-\bar{\delta}_{J,k})\big),$$

and $\mathcal{P}_J$ with $(\mathcal{P}_J)_{13} = (\mathcal{P}_J)_{31} = (\mathcal{P}_J)_{24} = (\mathcal{P}_J)_{42} = -g_J/2$, $(\mathcal{P}_J)_{14} = (\mathcal{P}_J)_{41} = -\delta_{J,k}/2$, $(\mathcal{P}_J)_{23} = (\mathcal{P}_J)_{32} = -\bar{\delta}_{J,k}/2$.

2 The transform $T = T_{\rm rot}\, T_{\rm sq}$

Phased sublattice rotation.

$$\alpha_k = \frac{e^{-i\varphi_k/2}\, b_{kA} + e^{+i\varphi_k/2}\, b_{kB}}{\sqrt{2}}, \qquad \beta_k = \frac{e^{-i\varphi_k/2}\, b_{kA} - e^{+i\varphi_k/2}\, b_{kB}}{\sqrt{2}}, \tag{1}$$

with $\varphi_k \to -\varphi_k$ for the $-k$ modes.

Two–mode squeeze on (α_k, α_{-k}) and (β_k, β_{-k}). With squeeze parameters

$$z_{k,\alpha} = \tfrac{1}{4}\ln\!\big(1 + 2\kappa_k\big), \qquad z_{k,\beta} = \tfrac{1}{4}\ln\!\big(1 - 2\kappa_k\big), \tag{2}$$

i.e., writing $x_\alpha = 1 + 2\kappa_k$, $x_\beta = 1 - 2\kappa_k$,

$$\cosh z_{k,s} = \tfrac{1}{2}\big(x_s^{1/4} + x_s^{-1/4}\big), \qquad \sinh z_{k,s} = \tfrac{1}{2}\big(x_s^{1/4} - x_s^{-1/4}\big).$$

The composite $T = T_{\rm rot}\, T_{\rm sq}$ is paraunitary, $T^\dagger \tau_3\, T = \tau_3$ with $\tau_3 = \mathrm{diag}(\mathbb{1}_4, -\mathbb{1}_4)$ (verified symbolically to the zero matrix).

3 Diagonalising H_C: the bands

$\tilde{\mathcal{H}}_C = T^\dagger \mathcal{H}_C\, T$ is diagonal, with all off–diagonal and pairing entries identically zero and physical band frequencies

$$\boxed{\omega_{\alpha,k} = \sqrt{1 + 2\kappa_k}\,, \qquad \omega_{\beta,k} = \sqrt{1 - 2\kappa_k}\,.} \tag{3}$$

4 Transforming H_J: the analytic coefficients

Write the transformed *physical* Josephson block as

$$2\, T^\dagger \mathcal{H}_J\, T = \begin{pmatrix} \tilde{\mathcal{A}}_J & \tilde{\mathcal{B}}_J \\ \tilde{\mathcal{B}}_J^* & \tilde{\mathcal{A}}_J^* \end{pmatrix},$$

in the quasiparticle basis $(\alpha_k, \beta_k, \alpha_{-k}, \beta_{-k})$; $\tilde{\mathcal{A}}_J$ is the number block and $\tilde{\mathcal{B}}_J$ the induced–pairing block. Introduce $w_\alpha = \omega_{\alpha,k}$, $w_\beta = \omega_{\beta,k}$ (the band amplitudes of (3)).

4.1 Structure ("Step 9b")

$\tilde{\mathcal{A}}_J$ is *momentum–block–diagonal* (it flips the band but never k) and $\tilde{\mathcal{B}}_J$ is *momentum–block–anti–diagonal* (every nonzero entry is a zero–momentum $\gamma_k\, \gamma_{-k}$ pair):

$$\tilde{\mathcal{A}}_J[\alpha_k, \alpha_{-k}] = \tilde{\mathcal{A}}_J[\alpha_k, \beta_{-k}] = 0, \qquad \tilde{\mathcal{B}}_J[\alpha_k, \alpha_k] = \tilde{\mathcal{B}}_J[\alpha_k, \beta_k] = 0. \tag{4}$$

4.2 Closed forms

Band–diagonal number shifts.

$$\tilde{\mathcal{A}}_J[\alpha_k, \alpha_k] = \frac{g_J}{w_\alpha} \;-\; w_\alpha\, \mathrm{Re}\big(e^{-i\varphi_k}\, \delta_{J,k}\big) = \frac{g_J}{w_\alpha} - \frac{w_\alpha}{2}\Big(e^{-i\varphi_k}\delta_{J,k} + e^{i\varphi_k}\bar{\delta}_{J,k}\Big), \tag{5}$$

$$\tilde{\mathcal{A}}_J[\beta_k, \beta_k] = \frac{g_J}{w_\beta} \;+\; w_\beta\, \mathrm{Re}\big(e^{-i\varphi_k}\, \delta_{J,k}\big). \tag{6}$$

Interband coupling (same k).

$$\boxed{\tilde{\mathcal{A}}_J[\alpha_k,\beta_k] = \frac{(1-4\kappa_k^2)^{1/4}}{2}\, e^{-i\varphi_k}\big(\delta_{J,k} - e^{2i\varphi_k}\,\bar{\delta}_{J,k}\big) = i\,\rho_k,} \tag{7}$$

where the amplitude ρ_k is *real*,

$$\boxed{\rho_k = (1-4\kappa_k^2)^{1/4}\,\mathrm{Im}\big(e^{-i\varphi_k}\,\delta_{J,k}\big) = (1-4\kappa_k^2)^{1/4}\,\frac{\big(g_{c2}g_{J1} - g_{c1}g_{J2}\big)\sin k}{\kappa_k}\,.} \tag{8}$$

The prefactor is $\sqrt{w_\alpha}\sqrt{w_\beta} = \big((1+2\kappa_k)(1-2\kappa_k)\big)^{1/4} = (1-4\kappa_k^2)^{1/4}$. $\tilde{\mathcal{A}}_J[\alpha_k,\beta_k]$ vanishes iff $\arg\delta_{J,k} = \varphi_k$ (Josephson and capacitive Bloch phases aligned) — in particular on $\chi_C = 0$.

Induced pairing (zero–momentum pairs).

$$\tilde{\mathcal{B}}_J[\alpha_k,\alpha_{-k}] = -\frac{g_J}{w_\alpha} \;-\; w_\alpha\,\mathrm{Re}\big(e^{-i\varphi_k}\,\delta_{J,k}\big), \tag{9}$$

$$\tilde{\mathcal{B}}_J[\beta_k,\beta_{-k}] = -\frac{g_J}{w_\beta} \;+\; w_\beta\,\mathrm{Re}\big(e^{-i\varphi_k}\,\delta_{J,k}\big), \tag{10}$$

$$\tilde{\mathcal{B}}_J[\alpha_k,\beta_{-k}] = \tilde{\mathcal{A}}_J[\alpha_k,\beta_k] \;=\; i\,\rho_k. \tag{11}$$

Readout. The band–diagonal number shift $\tilde{\mathcal{A}}_J[\gamma_k,\gamma_k]$ and the same–mode induced pairing $\tilde{\mathcal{B}}_J[\gamma_k,\gamma_{-k}]$ differ *only* in the sign of the g_J/w_γ term (compare (5) with (9), (6) with (10)). The interband number coupling and the α_k–β_{-k} zero–momentum pair carry the *identical* coefficient $i\rho_k$ ((7), (11)).

5 TLS truncation: the hybridised state

Cavity dressing (coupling γ, resonator frequency ω_r) couples $|\alpha_k,0\rangle$ to $|\beta_k,1\rangle$ through the interband coefficient (7). Truncating to that doublet,

$$H_{\mathrm{TLS}}(k) = \begin{pmatrix} \omega_{\alpha,k} & -\gamma\,\tilde{\mathcal{A}}_J[\alpha_k,\beta_k] \\ -\gamma\,\tilde{\mathcal{A}}_J[\alpha_k,\beta_k] & \omega_{\beta,k}+\omega_r \end{pmatrix} = \bar{E}_k\,\mathbb{1} + \frac{\Delta_k}{2}\,\sigma_z + \gamma\rho_k\,\sigma_y, \tag{12}$$

with $\bar{E}_k = \frac{1}{2}(\omega_{\alpha,k}+\omega_{\beta,k}+\omega_r)$ and $\Delta_k = \omega_{\alpha,k} - \omega_{\beta,k} - \omega_r$. Because $\tilde{\mathcal{A}}_J[\alpha_k,\beta_k] = i\rho_k$ is purely imaginary, the pseudo–spin $\boldsymbol{h}_k = \big(0,\ \gamma\rho_k,\ \Delta_k/2\big)$ is *planar* (no σ_x component). Its eigenvalues are

$$E_\pm(k) = \bar{E}_k \pm R_k, \qquad R_k = \sqrt{(\gamma\rho_k)^2 + (\Delta_k/2)^2}\,, \tag{13}$$

and the lower eigenvector, with polar angle $\theta_k = \mathrm{atan2}\big(\gamma\rho_k,\ \Delta_k/2\big)$, is

$$c_k = \big(i\sin\tfrac{\theta_k}{2},\ \cos\tfrac{\theta_k}{2}\big). \tag{14}$$

Lift to the chain / photon Hilbert space. With the SSH sublattice Bloch spinors $u_{\alpha,k} = (1,\ e^{-i\varphi_k})/\sqrt{2}$, $u_{\beta,k} = (1,\ -e^{-i\varphi_k})/\sqrt{2}$,

$$|\psi_k\rangle = i\sin\tfrac{\theta_k}{2}\;|u_{\alpha,k}\rangle\otimes|0\rangle \;+\; \cos\tfrac{\theta_k}{2}\;|u_{\beta,k}\rangle\otimes|1\rangle, \tag{15}$$

in the basis $\big((A,0),(B,0),(A,1),(B,1)\big)$, normalised for every k.

6 Analytical Zak phase of the hybridised band

Everything is now an explicit function of k through $\delta_{C,k} = g_{c1} + g_{c2}e^{-ik}$ and $\delta_{J,k} = g_{J1} + g_{J2}e^{-ik}$.

6.1 Energies and spinors, explicitly

$$\omega_{\alpha,k} = \sqrt{1+2\kappa_k}, \qquad \omega_{\beta,k} = \sqrt{1-2\kappa_k}, \qquad \Delta_k = \omega_{\alpha,k} - \omega_{\beta,k} - \omega_r, \tag{16}$$

$$\rho_k = (1-4\kappa_k^2)^{1/4}\,\frac{(g_{c2}g_{J1} - g_{c1}g_{J2})\sin k}{\kappa_k}, \qquad R_k = \sqrt{(\gamma\rho_k)^2 + \tfrac{1}{4}\Delta_k^2}, \quad E_\pm(k) = \bar{E}_k \pm R_k, \tag{17}$$

with $\bar{E}_k = \frac{1}{2}(\omega_{\alpha,k} + \omega_{\beta,k} + \omega_r)$ as in (12). The natural eigenvector (14) is *anti–periodic* across the BZ whenever θ_k winds an odd number of times. The periodic–gauge vector

$$\tilde{c}_k = e^{i\theta_k/2}\,c_k = \tfrac{1}{2}\Big(i\big(e^{i\theta_k}-1\big),\;\; e^{i\theta_k}+1\Big) \tag{18}$$

is a function of $e^{i\theta_k}$ alone, hence smooth and periodic. The hybridised state $|\psi_k\rangle$ is taken with the $\tilde{c}_k$ amplitudes.

6.2 Derivatives

Only two derivatives enter the Berry connection:

$$\varphi_k' = -\,\frac{g_{c2}\,(g_{c2} + g_{c1}\cos k)}{g_{c1}^2 + g_{c2}^2 + 2g_{c1}g_{c2}\cos k}\,, \tag{19}$$

and, with $X_k = \Delta_k/2$ and $Y_k = \gamma\rho_k$,

$$\theta_k' = \frac{X_k Y_k' - Y_k X_k'}{X_k^2 + Y_k^2} \tag{20}$$

(explicit but lengthy; used only under the BZ integral and numerically).

6.3 The Berry connection integrand, in blocks

The local Berry connection of the hybridised state decomposes exactly:

$$\boxed{\mathcal{A}(k) \;=\; i\,\langle\psi_k|\partial_k\psi_k\rangle \;=\; \underbrace{\tfrac{1}{2}\,\varphi_k'}_{\text{orbital (SSH)}} \;-\; \underbrace{\tfrac{1}{2}\,\theta_k'}_{\text{two–level}}\,.} \tag{21}$$

Two ways to see the split:

- **By physical origin.** Both SSH Bloch spinors carry the *same* orbital connection,

  $$i\,\langle u_{\alpha,k}|\partial_k u_{\alpha,k}\rangle = i\,\langle u_{\beta,k}|\partial_k u_{\beta,k}\rangle = \tfrac{1}{2}\varphi_k',$$

  so it factors out of the $|\tilde{c}_k|^2$–weighted superposition, while the periodic pseudo–spinor contributes $i\,\langle\tilde{c}_k|\partial_k\tilde{c}_k\rangle = -\frac{1}{2}\theta_k'$.

- **By sublattice.** The A–sublattice components of $|\psi_k\rangle$ give $-\frac{1}{4}\theta_k'$; the B–sublattice components (which carry $e^{-i\varphi_k}$) give $\frac{1}{2}\varphi_k' - \frac{1}{4}\theta_k'$.

6.4 The two block integrals

Orbital block (closed form).

$$\int_0^{2\pi} \frac{\varphi_k'}{2}\, dk = \begin{cases} 0, & g_{c1} > g_{c2}, \\ -\pi, & g_{c2} > g_{c1}, \end{cases} \;=\; -\pi\, w_C, \tag{22}$$

where $w_C \in \{0, 1\}$ is the SSH winding ($\delta_{C,k}$ traces a circle of radius g_{c2} about g_{c1}).

Two–level block (winding number).

$$\int_0^{2\pi} \theta_k'\, dk = 2\pi\, w_h, \qquad w_h = \mathrm{wind}_0\big(\gamma\rho_k,\ \Delta_k/2\big), \tag{23}$$

the number of times the planar vector $(\gamma\rho_k, \Delta_k/2)$ circles the origin as k crosses the BZ. `Integrate` cannot resolve this for generic parameters; w_h is read off numerically.

6.5 Result

$$\boxed{\gamma_{\mathrm{Zak}} = \int_0^{2\pi} \mathcal{A}(k)\, dk = -\pi\, w_C - \pi\, w_h \;\equiv\; \pi\,(w_h - w_C) \pmod{2\pi} \;\in\; \{0, \pi\}\,.} \tag{24}$$

The invariant is quantised. w_C is fixed by the SSH dimerisation; the cavity enters *only* through w_h: $\Delta_k = \omega_{\alpha,k} - \omega_{\beta,k} - \omega_r$ shifts rigidly with ω_r, so $w_h = 1$ (the vector encircles the origin) precisely while

$$\omega_r \in \Big[\ \min_k\big(\omega_{\alpha,k} - \omega_{\beta,k}\big),\ \max_k\big(\omega_{\alpha,k} - \omega_{\beta,k}\big)\ \Big] \tag{25}$$

— the interband–splitting window — and $w_h = 0$ outside it. The resonator therefore toggles γ_{Zak} by π on crossing the interband resonance, independently of γ (which only sets how sharp the transition in ω_r is).

7 Cross–checks

Coefficients vs `PeriodicChainSW` (§7 of the notebook). For each of eight parameter points — generic k, both TRIMs $k = 0, \pi$, and both dimerisation sides $g_{c1} \gtrless g_{c2}$, all with $g_{c1} + g_{c2} < \frac{1}{2}$ — `PeriodicChainSW.sector_nambu`(k) builds $\mathcal{H}_J$ and `diagonalize_cap_sector`(k) builds the same T; then $2\,T^\dagger \mathcal{H}_J\, T$ is compared entry–by–entry with (5)–(11). All differences $\max|\Delta\tilde{\mathcal{A}}_J|$, $\max|\Delta\tilde{\mathcal{B}}_J| \sim 10^{-15}$ (double precision round–off).

Operator–level (§6). $\mathcal{H}_C$, $\mathcal{H}_J$ are also built as second–quantised operators; the operator form of T is substituted, the result normal–ordered (Q3), and the same coefficients extracted with `Coefficient`. They match (7), (9), (11), (10).

Zak phase vs `dressed_zak_phase` (§8.4, §9.7). $|\gamma_{\mathrm{Zak}}|/\pi$ from (24) is compared with the fully cavity–dressed Wilson–loop Zak phase of the lower hybridised band, `PeriodicChainSW.dressed_zak_phase(1)` (photon cutoff 6, $N = 8$):

#	(g_{c1}, g_{c2})	regime	ω_r	γ	w_C	w_h	$\frac{\lvert\gamma_{\mathrm{Zak}}\rvert}{\pi} \stackrel{?}{=} \frac{\gamma_{\mathrm{dressed}}}{\pi}$
1	$(0.18, 0.11)$	trivial	0.30	0.03	0	1	$1 = 1$
2	$(0.18, 0.11)$	trivial	0.55	0.03	0	1	$1 = 1$
3	$(0.18, 0.11)$	trivial	0.80	0.03	0	0	$0 = 0$
4	$(0.11, 0.18)$	topological	0.30	0.03	1	1	$0 = 0$
5	$(0.11, 0.18)$	topological	0.80	0.03	1	0	$1 = 1$
6	$(0.18, 0.11)$	trivial	0.30	0.06	0	1	$1 = 1$
7	$(0.18, 0.11)$	trivial	0.80	0.06	0	0	$0 = 0$

($g_{J1} = 1$, $g_{J2} = -0.5$ throughout.) All rows agree.